

TrES-1b

R-2158

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-1_110520 · created 2026-08-12T23:24:17+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2455701.7644 +/- 0.00072 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.122 +/- 0.019
Transit depth (Rp/Rs)^2	1.49 +/- 0.46 [%]
Orbital Inclination (inc)	89.77 +/- 0.93
Airmass coefficient 1 (a1)	1.535 +/- 0.019
Airmass coefficient 2 (a2)	-0.1587 +/- 0.0028
Scatter in the residuals of the lightcurve fit is	1.15 %
Best Comparison Star	None
Optimal Aperture	3.5
Optimal Annulus	11.29
Transit Duration (day)	0.1017 +/- 0.0027

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.122 $\pm$ 0.019 vs 0.1358 (deviation 0.73 σ)
Duration fit vs published	0.1017 d vs 0.1042 d (deviation 0.91 σ)
Mid-time O–C	-1.5 min
Depth SNR	6.4
Residual scatter	1.15 %

Light curve

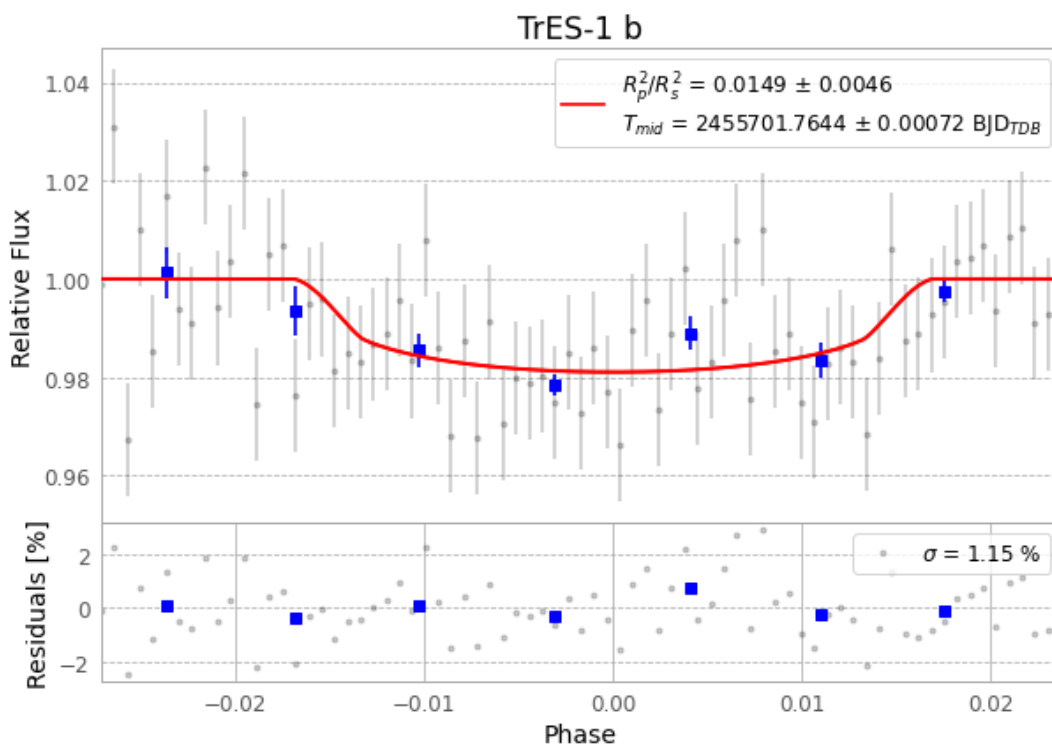


Figure 1 — FinalLightCurve_TrES-1 b_2011-05-19.png (SHA-256 0455112b89a10168...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [269, 204], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ee49e0ec68dd7c36...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

75 FITS frames (49 MB), TRES-1110520041813.FITS → TRES-1110520080013.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-110520.log (6a04afeb8de3...), FinalLightCurve_TrES-1 b_2011-05-19.png (0455112b89a1...), FinalLightCurve_TrES-1 b_2011-05-19.csv (881a43b0a6dd...)

Compute resources

Wall time 128.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-12 23:24Z.